# Application Data Sheet 37 CFR 1.76

The Application Data Sheet is part of the provisional or non-provisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.

## Inventor Information

# of inventors: 1

1. **Eric Haywood**

**Residence Information**
US Residency
Fort Collins, CO
UNITED STATES

**Mailing Address**
149 Remington Street Unit 425
Fort Collins, CO 80524
UNITED STATES

## Application Information

| | |
|---|---|
| Customer number | **221639 -** |
| Correspondence address | --- |
| Title of invention | **CLOUD-MEDIATED CROSS-DEVICE STATE SYNCHRONIZATION WITH PHI-INTERVAL HEARTBEAT AND DETERMINISTIC CONFLICT RESOLUTION** |
| Attorney docket number | **---** |
| Entity status | --- |

| | |
|---|---|
| Application type | Provisional Application under 35 USC 111(b) |
| Subject matter | Utility |
| Total number of drawing sheets | --- |
| Filing by reference | No |

## Representative Information

# of representatives: 0

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32).

Data was not provided for this section.

## Domestic Benefit/National Stage Information

# of benefit claims: 0

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c), 386(c), or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Data was not provided for this section.

## Foreign Priority Information

# of foreign priority claims: 0

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the Application Data Sheet constitutes the claim for priority as required by 35 U.S.C. 119 (b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a

copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Data was not provided for this section.

## Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

*Checking this box will cause the application to be examined under the first inventor to file provisions of the AIA.*

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 2016, 2013, will be examined under the first inventor to file provisions of the AIA.

## Authorization or Opt-Out of Authorization to Permit Access

When this Application Data Sheet is properly signed and filled with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

**NOTE:**
This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. **Priority Document Exchange (PDX)**
Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the China National Intellectual Property Administration (CNIPA), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its

related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h) (1).

B. **Search Results from U.S. Application to EPO**

Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with the search results from the instant application.

**NOTE:**

Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

## Applicant Information

# of applicants: 0

The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46.

Data was not provided for this section.

## Assignee Information including Non-Applicant Assignee Information

# of assignees: 1

An assignee-applicant identified in the "Applicant" section will appear on the patent application as an applicant.

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

**HeadySystems Inc.**
149 Remington Street Unit 425
Fort Collins, CO 80524
UNITED STATES

9702000327

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eric@headyconnection.org

**Organization**

## Signature

**NOTE:**

This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c)**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See CFR 1.4(d) for the manner of making signatures and certifications.

| Signature | First name | Last name | Registration # | Date |
|---|---|---|---|---|
| /Eric Haywood/ | ERIC | HAYWOOD | --- | 03/03/2026 |

UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

# ELECTRONIC ACKNOWLEDGEMENT RECEIPT

| APPLICATION # | RECEIPT DATE / TIME | ATTORNEY DOCKET # |
|---|---|---|
| **63/995,330** | **03/03/2026 02:19:32 PM Z ET** | **-** |

## Title of Invention

CLOUD-MEDIATED CROSS-DEVICE STATE SYNCHRONIZATION WITH PHI-INTERVAL HEARTBEAT AND DETERMINISTIC CONFLICT RESOLUTION

## Application Information

| | | | |
|---|---|---|---|
| APPLICATION TYPE | Utility - Provisional Application under 35 USC 111(b) | PATENT # | - |
| CONFIRMATION # | 6326 | FILED BY | ERIC HAYWOOD |
| PATENT CENTER # | 74697826 | FILING DATE | - |
| CUSTOMER # | 221639 | FIRST NAMED INVENTOR | Eric Haywood |
| CORRESPONDENCE ADDRESS | - | AUTHORIZED BY | - |

## Documents

**TOTAL DOCUMENTS: 2**

| DOCUMENT | PAGES | DESCRIPTION | SIZE (KB) |
|---|---|---|---|
| generatedADS74697826.pdf | 5 | Application Data Sheet | 109 KB |
| HS-2026-010-Final-Assigned-Updated-APP.TEXT.docx | 4 | Application body structured text document | 39 KB |

Warning: One or more pages are missing page numbering. Page numbering will be automatically applied after submission.

## Digest

| DOCUMENT | MESSAGE DIGEST(SHA-512) |
|---|---|
| generatedADS74697826.pdf | 345EF0C50A1CF33B0CA7E49A05B4762ADB393547792F9A4E1 |

A983AF54657E4D1C7B5C54C64DA25C88FEBCFFB2F09A22D2
05F7146366708CE5D370C3F4ADCF361

HS-2026-010-Final-Assigned-Updated-APP.TEXT.docx

A450F775C21FF4FCDC77D19620A53F09771D2BF882C120F01
EA93C5AAADE2139B5C2BE0E7379BB00767BBCDD38D907C4
716989766C5BE5042CDCD4725D512C76

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

**New Applications Under 35 U.S.C. 111**

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

**National Stage of an International Application under 35 U.S.C. 371**

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

**New International Application Filed with the USPTO as a Receiving Office**

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PROVISIONAL PATENT APPLICATION

Under 35 U.S.C. § 111(b)

# CLOUD-MEDIATED CROSS-DEVICE STATE SYNCHRONIZATION WITH PHI-INTERVAL HEARTBEAT AND DETERMINISTIC CONFLICT RESOLUTION

| | |
|---|---|
| Title of the Invention | Cloud-Mediated Cross-Device State Synchronization with Phi-Interval Heartbeat and Deterministic Conflict Resolution |
| Docket Number | HS-2026-010 |
| Inventor(s) | Eric Haywood |
| Residence(s) | 149 Remington Street Unit 425, Fort Collins, CO 80524 |
| Correspondence Address | HeadySystems Inc., 149 Remington Street Unit 425, Fort Collins, CO 80524 |
| Attorney/Agent (Name & Reg. No.) | None |
| U.S. Government Interest (Yes/No) | No |
| Gov. Agency & Contract No. | N/A |

## Title of the Invention

[0001] Cloud-Mediated Cross-Device State Synchronization with Phi-Interval Heartbeat and Deterministic Conflict Resolution

## Cross-Reference to Related Applications

[0002] This application claims the benefit of priority and is related to co-pending provisional applications filed concurrently herewith by the same inventor and assignee.

## Field of the Invention

[0003] This invention relates to systems and methods for synchronizing application state across multiple heterogeneous devices through a cloud-mediated sync mesh, using deterministic conflict resolution, phi-proportional heartbeat intervals and automatic authorization propagation.

## Background of the Invention

[0004] Cross-device synchronization is a longstanding problem in distributed systems. Existing solutions synchronize file-level data but do not provide real-time application state synchronization including authentication tokens, active module states, chat histories and filesystem bookmarks; automatic propagation of authorization grants; deterministic conflict resolution using cryptographic sync identifiers; or phi-proportional heartbeat intervals that produce naturally distributed network traffic patterns.

## Summary of the Invention

[0005] The invention provides a system and method as described herein.

[0006] Device Registry: A cloud-hosted registry that maintains metadata for all provisioned devices, uses heartbeat-based liveness detection and automatically prunes stale devices.

[0007] Sync State Manager: Maintains the canonical application state, supporting partial updates from any device with deterministic conflict resolution via SHA-256-based sync identifiers.

[0008] Authorization Propagation: When one device receives root authorization, the authorization is instantly propagated to all registered devices, so a single authorization authorizes the entire device fleet.

[0009] Phi-Proportional Heartbeat: Device heartbeats run on a phi-scaled interval (base interval × φ), producing naturally spaced network traffic that minimizes bandwidth contention during concurrent device synchronization.

[0010] Multi-Channel Sync: Synchronizes at least five distinct data channels — authentication state, module installation states, chat message history, user preferences and filesystem bookmarks — through a unified sync protocol.

## Detailed Description of the Preferred Embodiment

A. DeviceRegistry

[0011] The DeviceRegistry component stores a mapping of device identifiers to device records. Each record contains the device type, registration timestamp, last heartbeat timestamp, current sync state and other metadata. Heartbeat processing updates the last-seen timestamp and transitions the device state to connected. A listActive function returns devices that have sent a heartbeat within a configurable threshold (default 60 seconds), enabling stale device pruning.

B. SyncStateManager

[0012] The SyncStateManager maintains a versioned canonical state object containing fields such as version identifier, sync ID, root authorization state, module installation states, user preferences, chat history and filesystem bookmarks. Each state mutation generates a new sync ID by computing a SHA-256 hash of the serialized state concatenated with the current timestamp, truncated for compactness. Updates are applied using deep merge with field-level last-writer-wins semantics.

[0013] The last-writer-wins approach resolves conflicts by applying the update with the most recent timestamp; ties may be broken deterministically using string comparisons or hashed values. This deterministic conflict resolution ensures that all devices converge to the same canonical state even under concurrent updates.

C. CrossDeviceSync Orchestrator

[0014] The orchestrator coordinates device registration, state updates and broadcasts. On device registration, it pushes the full canonical state to the new device. When a state update arrives from any device, it applies the patch to the canonical state and broadcasts the updated state to all other active devices, excluding the source to prevent echo loops.

[0015] Authorization propagation ensures that when any device obtains an authorization grant (such as OAuth or access token), the updated authorization state is broadcast to every device in the mesh. Similarly, module installation updates, chat messages and preference changes are propagated across devices with sub-second latency. All sync events are emitted to a global event bus for external system integration.

## Claims

1. A computer-implemented method for cross-device state synchronization, comprising: (a) maintaining a cloud-hosted device registry with heartbeat-based liveness detection; (b) managing a canonical application state that is updated by partial patches from any registered device using deterministic conflict resolution via cryptographic sync identifiers; (c) upon receiving an authorization grant on any single device, automatically propagating the authorization to all other registered devices in the sync mesh; (d) broadcasting state updates from any source device to all other active devices while excluding the source device to prevent echo feedback; and (e) synchronizing at least authentication state, module installation states, chat message history, user preferences and filesystem bookmarks through a unified protocol.

2. The method of claim 1, wherein the deterministic conflict resolution comprises generating a sync identifier by computing a SHA-256 hash of the serialized canonical state concatenated with a monotonic timestamp and applying updates using field-level last-writer-wins merge semantics.

3. The method of claim 1, wherein the device heartbeat interval is derived from the golden ratio ($\varphi \approx 1.618$) applied to a base interval, producing naturally distributed network traffic patterns that minimize bandwidth contention during concurrent device synchronization.

4. A cross-device synchronization system comprising: a device registry maintaining liveness-monitored device records; a sync state manager maintaining canonical application state with cryptographic conflict resolution; an authorization propagation engine that broadcasts auth grants across the device mesh; a broadcast engine that distributes state patches to all active devices excluding the update source; and a multi-channel sync protocol handling authentication, module states, chat history, preferences and filesystem bookmarks.

## Implementation Reference

[0016] The Cross-Device Sync service may be implemented in the file `src/services/cross-device-sync.js` using classes `DeviceRegistry`, `SyncStateManager` and `CrossDeviceSync`. The service depends on `heady-principles.js` for the PHI constant to compute heartbeat intervals.

## References

[1] Scheduling events at golden-ratio intervals ($\varphi \times 60$ minutes $\approx 37.1$ minutes) yields nearly uniform coverage and allows adding events while keeping coverage close to uniform 【10089563345220†L41-L50】

[2] The golden ratio has the slowest-converging continued fraction and is considered the most irrational number, minimizing resonance when used for proportional splits 【577248341085003†L203-L218】

[3] In distributed systems, last-writer-wins conflict resolution selects the write with the latest timestamp; tie-breakers use deterministic criteria【554987348588856†L818-L835】